

Process Compass

A stage-by-stage field guide to coffee leaf processing — from harvest to final moisture reduction. Each stage documents the key parameters, decision points, pitfalls, and the reasoning behind each choice.

Compiled and editorially structured by Citane / KoffyKraft · Thumpassery Estate, Karavallur, Kollam, Kerala · Source research belongs to original authors. Traditional knowledge belongs to communities of origin. Unpublished adaptations are labelled as such throughout.

- Published Research — peer-reviewed, cited
- Traditional Practice — documented, not formally validated
- Citane Adaptation — unpublished, from estate observation

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Harvest

Timing · Maturity · Handling

TIMING AND MATURITY

Leaf maturity at harvest is the single most consequential decision in the entire process. It determines which volatile compounds are available, which enzymes are active, and which processing pathways are possible downstream.

Parameter	Value / Range	Why it matters
Young flush (1–3 months)	Highest mangiferin, hexanal, (E,Z)-2,6-nonadienal · Low tannin	Green, refreshing, floral profiles possible · Fragile under heat
Mid-maturity (3–5 months)	Balanced chlorogenic acids · Moderate β -ionone precursors	Widest process range · Oolong, white, yeast ferment
Mature (5–7 months)	High tannin, high lipid · Low green volatiles	Black, roast, smoke, decoction · More structurally robust
Post-mature / yellow	Degraded chlorophyll · Elevated β -carotene breakdown products	Honey-ionone pathway only · Not suitable for green style

HARVEST PROTOCOL

→ Harvest in early morning — leaf temperature lower, enzyme activity slower, reduced risk of field oxidation before processing begins.

- Pick individual leaves or small clusters — avoid stripping branches which bruises retained leaves
- Count from branch tip: 1st–3rd leaf = young flush (highest volatile potential); 5th–7th = mature
- Handle cut surface upward where possible — exposed cell wall at cut point begins oxidising immediately
- Do not stack harvested leaves in deep piles in the field — compression bruising initiates enzyme cascade
- KoffyKraft observation: leaves harvested from rubber-shaded Chandragiri Arabica at 130m show slower enzyme activation post-harvest than open-canopy leaves — likely due to lower ambient leaf temperature

■ Do not harvest wet leaves (after rain or heavy dew) for green or oolong style — surface moisture accelerates unwanted fermentation before enzyme arrest can be achieved.

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Transfer and Holding

Container · Atmosphere · Time window

The transfer window — time between harvest and first processing step — is an active period. Enzymes begin working the moment the leaf is cut. Managing this window determines whether you arrive at the next stage with the chemistry you intended.

IF	THEN
Target: green / refreshing profile	Process within 30 min of harvest. No holding. Steaming must begin immediately.
Target: oolong / floral / sweet	Allow 0–2h ambient hold. This is the beginning of the wither window.
Target: black / woody / decoction	Hold up to 4h acceptable. Mature leaves are more tolerant.
Unavoidable delay (>2h for green target)	Refrigerate at 4–8°C. Cold holds enzyme activity. Resume within 6h.

CONTAINER REQUIREMENTS

- Breathable containers (bamboo baskets, open crates) — prevents CO₂ and heat build-up from enzymatic activity
- Avoid sealed plastic bags for holding >30 min — anaerobic atmosphere initiates unwanted fermentation
- Single layer preferred for young flush — stacking causes compression bruising and uneven enzyme activation
- For longer holding (oolong / black target): loose layer in shallow trays at ambient temperature is acceptable
- Traditional Engere practice: leaves pounded then immediately decocted — zero transfer window observed (Yohannis et al. 2026)

■ Sealed containers + warm ambient temperature (Kerala 28–32°C) = uncontrolled fermentation within 1–2h. If this is not the intended process, it is a critical quality loss point.

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Surface Preparation

Wash decision · Contamination · Compound integrity

Whether to wash — and how — is a genuine decision with compound-level consequences. Water contact initiates cell surface reactions. The correct protocol depends on what comes next.

IF	THEN
Light field dust only, green target	Light brush or dry wipe only. No water.
Moderate contamination, green target	Brief mist rinse (spray bottle). Pat dry immediately. Do not soak.
Oolong / black / ferment target	Quick rinse acceptable. Excess moisture must be removed before wither.
Decoction target (Engere / Chemo style)	Rinse acceptable — compounds extracted in brew water; surface moisture not critical.
Heavily contaminated leaves	Redirect to secondary batch for quick-processed or decoction profile. Do not use for premium green or oolong.

■ Soaking — even briefly — causes rapid intracellular damage through osmotic pressure. Green volatile compounds (hexanal, (E,Z)-2,6-nonadienal) are destroyed before processing begins. Never soak leaves destined for green or refreshing profiles. (● Steger et al. 2022 — confirmed soaking as a quality loss pathway)

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Wither

Time · Temperature · Atmosphere · Target profile

Withering is the first intentional transformation stage. It controls the extent of enzymatic oxidation before any heat treatment. Duration and temperature together determine which volatile families develop and which are suppressed. This is where process

style is committed.

Parameter	Value / Range	Why it matters
Green style (JGTP/CGTP)	0h — immediate enzyme arrest	Preserves hexanal, (E,Z)-2,6-nonadienal. No withering at all.
White / sun-dried style	12–24h, ambient, sun or shade	Slow enzymatic conversion. β -ionone and α -ionone precursors develop. Floral, sweet potential.
Oolong style (OTP)	12–24h, 22–25°C, shade or indoor	Partial oxidation window. OCLT compounds develop (limonene, phenyl-methylene heptanoate, diphenyl sulfone). Optimal: 24h (Fibrianto et al. 2025).
Black style (BTP)	24h, ambient	Full oxidation commitment. Tannin polymerisation proceeds. Woody, earthy character emerges.
Yeast fermentation path	12–18h wither before inoculation	Primes leaf substrate. Partial enzyme activation before anaerobic switch.
Shade wither (■ Citane)	12–24h, low light, 22–25°C	Slower than sun wither. More controlled ionone development without UV-driven carotenoid degradation.

WITHER ATMOSPHERE

- Indoor shade wither: most controllable. Consistent temp, no direct UV, manageable humidity.
- Sun wither: accelerates carotenoid conversion (→ β -ionone). Higher sweet potential but harder to control. Stop before leaves become crisp.
- Shade wither at Thumpassery: rubber canopy provides natural diffused light environment. Ambient 22–28°C. Slower but consistent.
- Avoid wither in sealed or low-ventilation spaces — CO■ from leaf respiration can trigger unintended anaerobic conditions.
- Target moisture loss during wither: approximately 20–30% of fresh weight. Leaves should be pliable but not brittle.

■ Withering above 30°C accelerates oxidation past the intended window — oolong-target leaves can enter black-style oxidation within 6–8h instead of 24h. Monitor ambient temperature, especially in Kerala monsoon season when night temperatures remain high.

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Manipulation

Rolling · Wring · Cell rupture · Post-wring hold

Mechanical manipulation controls cell wall rupture — the release of intracellular compounds into contact with oxygen and each other. This is a precise intervention: too little and the substrate is not primed; too much and the oxidative cascade runs ahead of the intended profile.

Parameter	Value / Range	Why it matters
Light roll (10–15% rupture estimate)	Oolong / floral / fruity target	Releases ionone precursors without triggering full oxidative cascade. Moisture gradient created.
Medium wring (30–40% rupture)	Black / woody target	Significant tannin-enzyme contact. Full oxidation now proceeding. Commit to black profile.
Blend / cut (green target)	Green / refreshing target	Hexanal and 4-heptenal released. Immediately followed by enzyme arrest (steam or oven).
No manipulation	White / sun-dried / decoction	Whole leaf drying. Minimal intracellular disruption. Flavour develops through surface chemistry only.

WRING HOLD AND POST-WRING TEMPERATURE

Wringing generates a small but measurable temperature increase in the leaf mass from mechanical energy. This post-wring temperature rise accelerates enzyme activity — intentionally or not.

- Observe post-wring leaf temperature before committing to hold time. A 2–3°C rise is normal.
- If post-wring temp exceeds 28°C in ambient conditions: shorten hold or move to cooler space to slow cascade.

- Wring hold (leaves held in wrung/rolled position): 15–30 min allows oxidative reaction to develop from rupture points while limiting how far it progresses.
- Releasing and spreading leaves after hold: resumes normal surface oxidation at slower rate. Use this to fine-tune before next step.
- The wringing-to-hold-to-release sequence mirrors oolong tea processing principles (Chen & Kitts 2018 — phytochemical effects of processing).

Note: All wring/roll intensity descriptions are observational estimates. Formal quantification of % cell rupture requires microscopy. These are working field descriptors, not validated measurements. Label as Citane Adaptation (unpublished).

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Reaction Window

Let run · Stall · Stop · Fermentation cascades

After manipulation, the leaf is in an active reactive state. The question is not whether reactions are happening — they always are — but whether to let them run, stall them at a chosen point, or stop them. Each choice produces a different flavour outcome.

IF	THEN
Stop immediately (green target)	Steam at 100°C, 2–3 min. Full enzyme arrest. Locks green volatile profile in place.
Stall at partial oxidation (oolong)	Monitor colour change (green → yellow-green edge). Then dry at 70°C. Locks oolong profile.
Let run to completion (black)	Allow full 24h wither. No interruption. Oven dry when leaves are copper-red.
Introduce yeast fermentation	Apply <i>Saccharomyces cerevisiae</i> starter (40mL per batch). Anaerobic, 25°C, 12h. Then dry. (● Steger et al. 2022)
Wild fermentation	No starter. Anaerobic hold, 25°C, 12h. Variable outcomes — document each batch. (● Steger et al. 2022)
Koji fermentation (■ Citane)	Koji inoculation (<i>Aspergillus oryzae</i>). Humidity-controlled environment. Enzymatic pathway distinct from yeast. Theoretical — not yet formally documented for CLT.
Water soak activation (■ Citane)	Brief controlled soak (15–30 min, ambient temp) to activate surface enzymes then immediate arrest. Theoretical — requires testing.
Koji mist (■ Citane)	Light spray of Koji spore suspension on leaf surface. Partial surface inoculation. Theoretical pathway.

ENZYME TEMPERATURE REFERENCE

Enzyme	Temperature range	Relevance to CLT process
Polyphenol oxidase (PPO)	Active: 15–35°C · Inactivated: >70°C	Primary oxidation enzyme. Controls green → black conversion.
Peroxidase	Active: 20–40°C · Inactivated: >80°C	Secondary oxidation. More heat-resistant than PPO.
Lipoxygenase	Active: 15–30°C · Inactivated: >60°C	Produces green aldehydes (hexanal, etc). Inactivates at lower temp — steaming effective.
Protease / amylase	Active: 20–50°C	Amino acid and sugar release. Active during fermentation window.

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Moisture Reduction

Method · Temperature · Duration · Target moisture

Moisture reduction serves two purposes: it arrests remaining enzyme activity through heat or desiccation, and it stabilises the leaf for storage and brewing. The method chosen here affects volatile retention — high heat drives off aromatic compounds faster than low heat.

Parameter	Value / Range	Why it matters
Oven dry	70°C / 4h	Standard for green, oolong, black. Documented in Fibrianto et al. 2025 and Steger et al. 2022. Reliable enzyme arrest without excessive volatile loss.
Sun dry	≥48h ambient	White style and traditional pathways. Slow moisture loss preserves floral ionones. Weather-dependent — humidity risk.
Flat-pan roast (Kuti)	High heat, 15–20 min	Simultaneous drying and roast character development. Mature leaves only. Pyrazines and guaiacol formed.
Smoke-dry (Kawa Daun)	2–4h, closed furnace, cassia wood	Simultaneous drying and smoke infusion. Cinnamomum burmannii volatile oils absorbed into leaf.
Air dry (low humidity)	24–48h at 25°C with airflow	Gentler than oven. Suitable for wild fermentation outcomes where high heat may destroy fermentation-derived esters.
Koji / ferment path	70°C / 4h after fermentation	Heat arrest of yeast or Koji activity. Then dry fully. Same oven protocol as standard.

TARGET MOISTURE

- Target moisture content for storage: 5–8% (crisp, not powder). Leaves should snap cleanly, not bend.
- Over-drying (>4h at 70°C): volatile compounds continue to evaporate. Aromatic intensity decreases.
- Under-drying: residual moisture enables mould growth in storage within days. Do not skip this step.
- No formal moisture measurement equipment required at small scale — snap test and visual assessment sufficient for field use.



QC Checkpoints

Brix · pH · Sensory · Documentation

Optional measurement points that can help track process consistency across batches. None of these are mandatory at small scale — sensory assessment is the primary tool. Introduce instruments only where they add information that sensory cannot provide.

Checkpoint	When	What it tells you
Brix (refractometer)	After wither (before drying)	Tracks sugar concentration increase from moisture loss. Useful for monitoring wither consistency across batches. Target Brix change: +2–4° over 24h wither (■ Citane estimate — not formally documented for CLT).
pH (meter or strips)	After fermentation · After brew	Fermentation lowers pH as organic acids develop. Brew pH affects mouthfeel perception. CLT brew pH documented at 5.57–7.06 depending on altitude and variety (● Fibrianto et al. 2024).
Sensory — colour	After wither · After drying	Colour is the most reliable visual process marker. Green → yellow-green (oolong window) → copper-red (black) → dark brown (roast). Colour assessment requires consistent lighting.
Sensory — aroma	After wither · After drying · After brew	Primary quality signal. Green aldehydes (grassy, cut-grass) should be present for green target. Floral-sweet for oolong. Check for off-notes: sour, musty, acrid = process failure.
Batch documentation	All stages	Harvest date and time, leaf maturity, wither start/end time and temp, manipulation type, fermentation duration if used, drying method and duration, final sensory notes. Repeat success requires documentation.